POINTWISE DIAGONAL COHOMOLOGICAL VANISHING IMPLIES COMPLETENESS OF THE POSTNIKOV TOWER

THE STACKS PROJECT AUTHORS

Editorial note. The bound n(x) is chosen separately for each fixed cohomological degree m; no global bound on cohomological dimension is assumed.

Editorial note. The stated canonical comparison is understood in the standard compatible-map sense of the cited derived-limit remark; the proof shows every compatible choice is an isomorphism.

Editorial scope. One original target; complete source-local proof chain included, standard derived-sheaf primitives explicitly delimited.

Allowed standard prerequisites (editorial).

- Derived inverse limits and Milnor sequence. RΓ(U,-) commutes with derived countable inverse limits; its cohomology lies in the short exact sequence with lim and R^1lim. Eventually constant inverse systems have R^1lim=0.
- Standard truncation triangles and sheafification. Canonical truncations fit in the cohomology-object triangle, and H^m(K) is the sheaf associated to U↦H^m(U,K).

Source reference. Original section 0H72: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 10557–10586. Standard definition of the compatible map to the truncation inverse limit. Displayed pointer is the original stable Stacks tag.

Source reference. Original section 08J5: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 4075–4109. Standard truncation triangle; it does not establish the pointwise vanishing criterion. Displayed pointer is the original stable Stacks tag.

Source reference. Original lemma 0D60: focused-seeds/ACL/sources/stacks-snapshot-20261011/cohomology.tex, source lines 9456–9467. Explicit permitted Milnor exact sequence. Displayed pointer is the original stable Stacks tag.

Source reference. Original lemma 0BKJ: focused-seeds/ACL/sources/stacks-snapshot-20261011/cohomology.tex, source lines 7875–7883. Standard description of cohomology sheaves, used only to choose and shrink representatives. Displayed pointer is the original stable Stacks tag.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $E \in D(\mathcal{O}_X)$. Assume that for every $x \in X$ there exist a function $p(x, -) : \mathbf{Z} \to \mathbf{Z}$ and a fundamental system $\mathfrak{U}_x$ of open neighbourhoods of x such that*

$$H^p(U, H^{m-p}(E)) = 0 \text{ for } U \in \mathfrak{U}_x \text{ and } p > p(x, m)$$

Then the map $E \to R\lim \tau_{\geq -n} E$ of Derived Categories, Remark 0H72 is an isomorphism in $D(\mathcal{O}_X)$.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.2. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let (K_n) be an inverse system in $D(\mathcal{O}_X)$. Let $x \in X$ and $m \in \mathbf{Z}$. Assume there exist an integer $n(x)$ and a fundamental system $\mathfrak{U}_x$ of open neighbourhoods of x such that for $U \in \mathfrak{U}_x$*

(1) *$R^1 \lim H^{m-1}(U, K_n) = 0$, and*
(2) *$H^m(U, K_n) \to H^m(U, K_{n(x)})$ is injective for $n \geq n(x)$.*

Then the map on stalks $H^m(R \lim K_n)_x \to H^m(K_{n(x)})_x$ is injective.

Proof. Let γ be an element of $H^m(R \lim K_n)_x$ which maps to zero in $H^m(K_{n(x)})_x$. Since $H^m(R \lim K_n)$ is the sheafification of $U \mapsto H^m(U, R \lim K_n)$ (by Lemma 0BKJ) we can choose $U \in \mathfrak{U}_x$ and an element $\tilde{\gamma} \in H^m(U, R \lim K_n)$ mapping to γ. Then $\tilde{\gamma}$ maps to $\tilde{\gamma}_{n(x)} \in H^m(U, K_{n(x)})$. Using that $H^m(K_{n(x)})$ is the sheafification of $U \mapsto H^m(U, K_{n(x)})$ (by Lemma 0BKJ again) we see that after shrinking U we may assume that $\tilde{\gamma}_{n(x)} = 0$. For this U we consider the short exact sequence

$$0 \to R^1 \lim H^{m-1}(U, K_n) \to H^m(U, R \lim K_n) \to \lim H^m(U, K_n) \to 0$$

of Lemma 0D60. By assumption (1) the group on the left is zero and by assumption (2) the group on the right maps injectively into $H^m(U, K_{n(x)})$. We conclude $\tilde{\gamma} = 0$ and hence $\gamma = 0$ as desired. □

Proof. Set $K_n = \tau_{\geq -n} E$ and $K = R \lim K_n$. The canonical map $E \to K$ comes from the canonical maps $E \to K_n = \tau_{\geq -n} E$. We have to show that $E \to K$ induces an isomorphism $H^m(E) \to H^m(K)$ of cohomology sheaves. In the rest of the proof we fix m. If $n \geq -m$, then the map $E \to \tau_{\geq -n} E = K_n$ induces an isomorphism $H^m(E) \to H^m(K_n)$. To finish the proof it suffices to show that for every $x \in X$ there exists an integer $n(x) \geq -m$ such that the map $H^m(K)_x \to H^m(K_{n(x)})_x$ is injective. Namely, then the composition

$$H^m(E)_x \to H^m(K)_x \to H^m(K_{n(x)})_x$$

is a bijection and the second arrow is injective, hence the first arrow is bijective. Set

$$n(x) = 1 + \max\{-m, p(x, m-1) - m, -1 + p(x, m) - m, -2 + p(x, m+1) - m\}.$$

so that in any case $n(x) \geq -m$. Claim: the maps

$$H^{m-1}(U, K_{n+1}) \to H^{m-1}(U, K_n) \quad \text{and} \quad H^m(U, K_{n+1}) \to H^m(U, K_n)$$

are isomorphisms for $n \geq n(x)$ and $U \in \mathfrak{U}_x$. The claim implies conditions (1) and (2) of Lemma 0.2 are satisfied and hence implies the desired injectivity. Recall (Derived Categories, Remark 08J5) that we have distinguished triangles

$$H^{-n-1}(E)[n+1] \to K_{n+1} \to K_n \to H^{-n-1}(E)[n+2]$$

Looking at the associated long exact cohomology sequence the claim follows if

$$H^{m+n}(U, H^{-n-1}(E)), \quad H^{m+n+1}(U, H^{-n-1}(E)), \quad H^{m+n+2}(U, H^{-n-1}(E))$$

are zero for $n \geq n(x)$ and $U \in \mathfrak{U}_x$. This follows from our choice of $n(x)$ and the assumption in the lemma. □